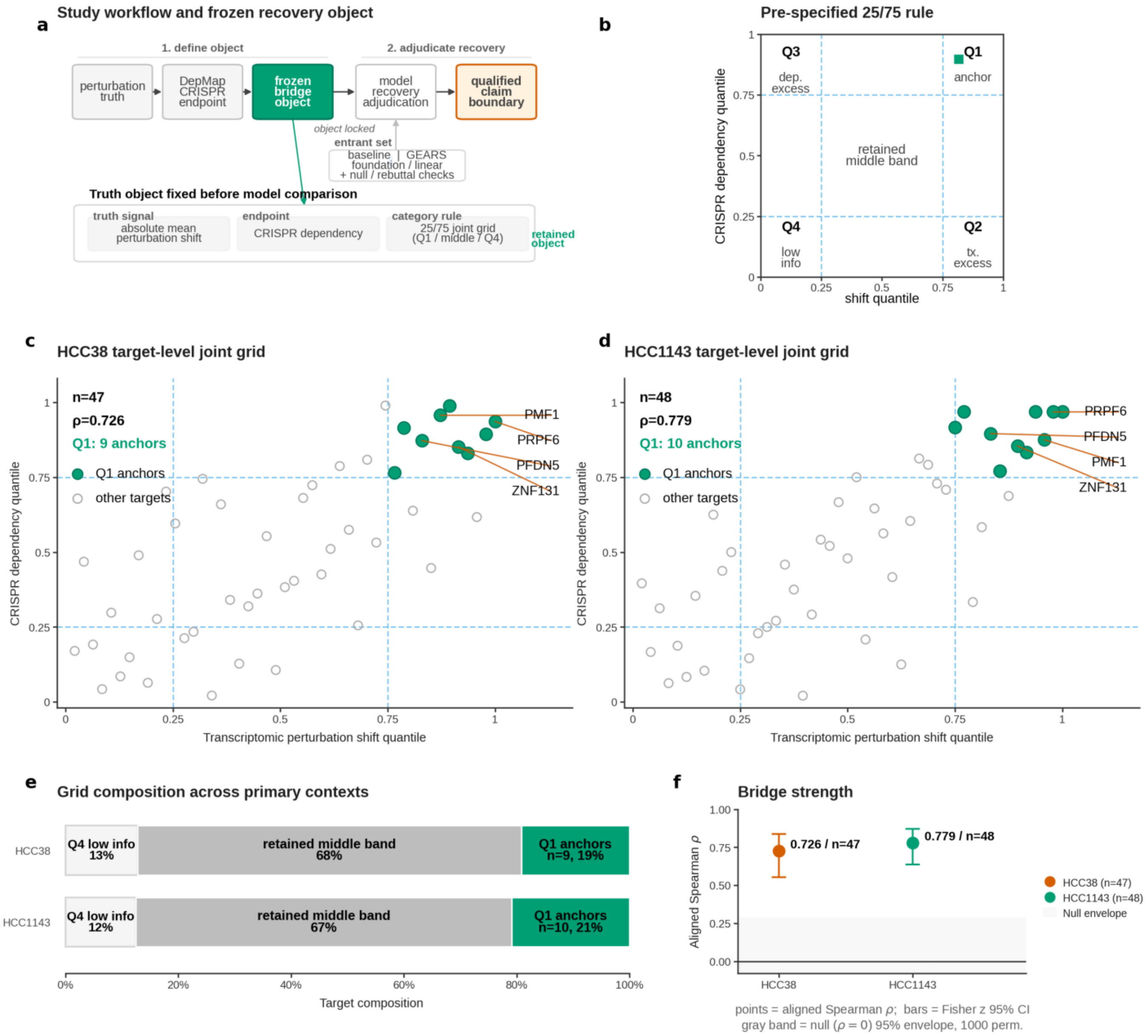


Fig. 1 | Framework element 1: A pre-specified perturbation-to-dependency bridge defines the truth object in HCC38 and HCC1143

a, Study workflow and frozen recovery object. The framework defines a phenotype-aligned truth object by aligning absolute mean perturbation shift (transcriptomic truth) to CRISPR DepMap dependency (fitness endpoint) before any model-scoring step. b, Pre-specified 25/75 category rule. Targets were classified on a pre-specified joint grid using the 25th and 75th percentiles of transcriptomic perturbation shift and CRISPR dependency, with corner-defined categories and a retained middle band. c, HCC38 target-level joint grid. Axes show within-context rank percentiles (shift_quantile and depmap_quantile), with n, aligned Spearman rho, and Q1 anchor count annotated in the panel. d, HCC1143 target-level joint grid. Axes show within-context rank percentiles, with n, aligned Spearman rho, and Q1 anchor count annotated in the panel. e, Grid composition across the two primary contexts. Q2 transcriptomic-excess and Q3 dependency-excess targets were not observed in either primary context and are retained as zero-count categories in the composition summary. f, Bridge strength summary.

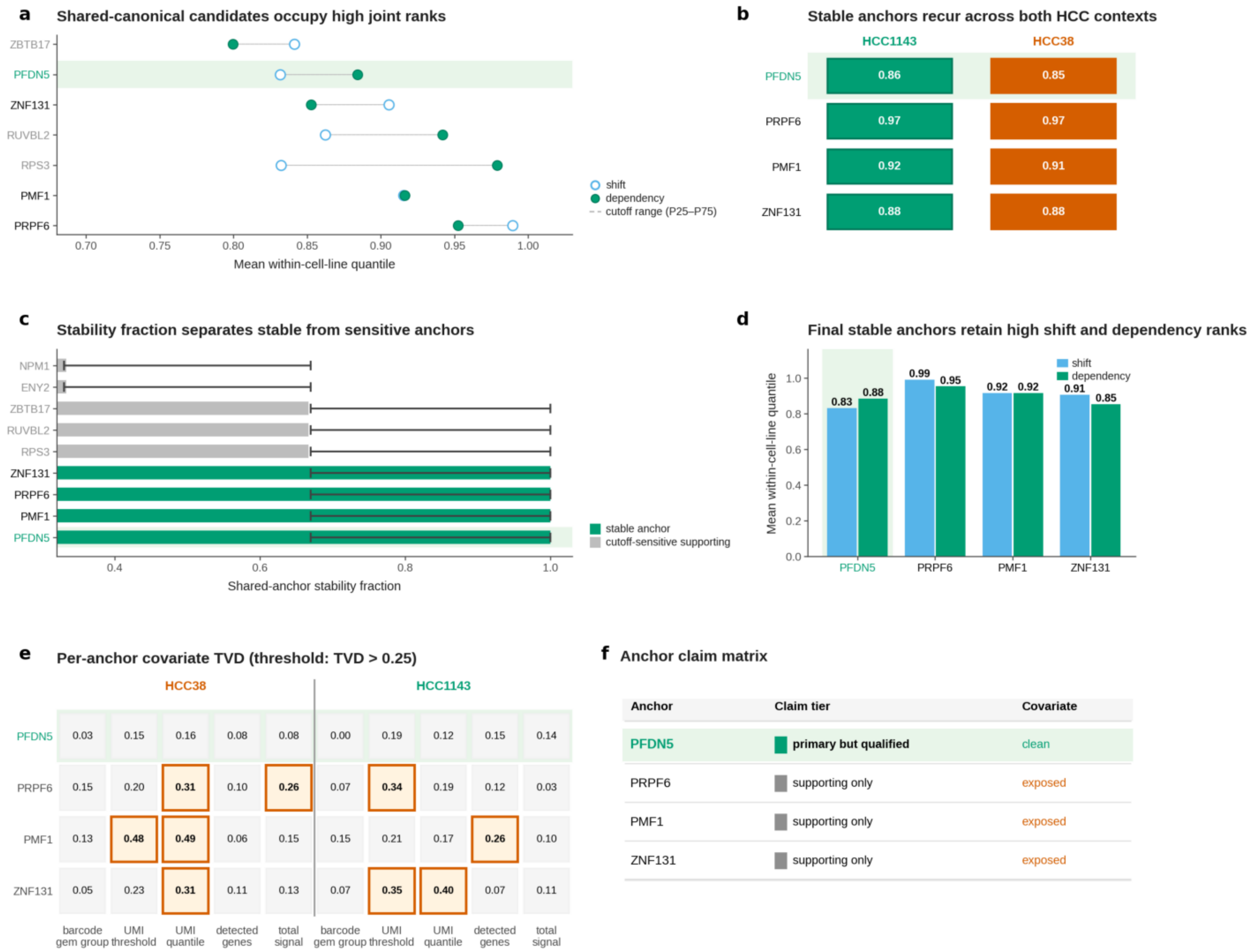
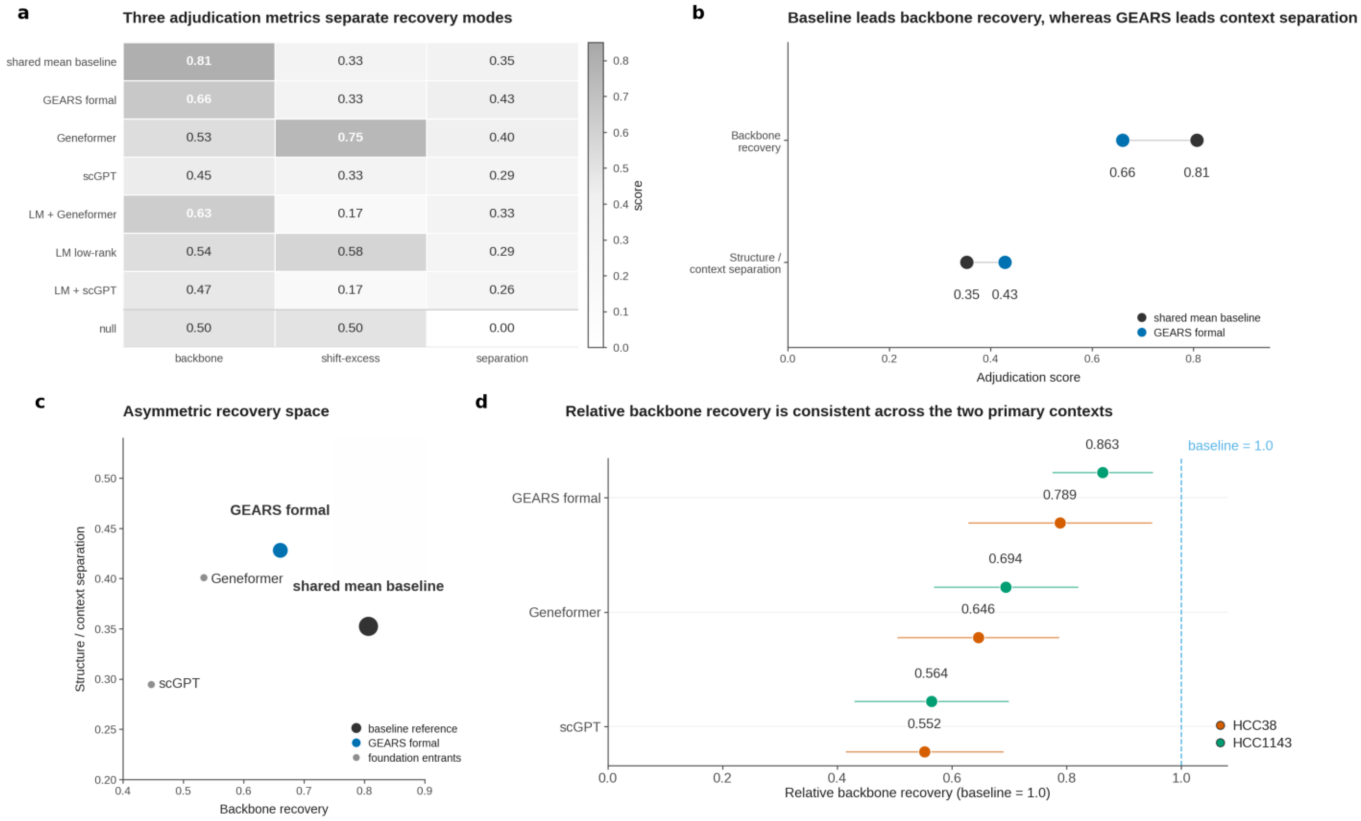


Fig. 2 | Framework element 2: Covariate-aware anchor tiering separates recurrent bridge support from target proof

a, Shared-canonical anchor ranking. Paired shift (open) and dependency (filled) within-cell-line quantile means for the seven shared-canonical candidates, ordered by the mean within-cell-line joint quantile (mean of shift and dependency quantile means). b, Low-ink recurrence matrix across the two primary HCC contexts (HCC1143 and HCC38) for the four final stable anchors. c, Stability fraction across stable and cutoff-sensitive shared-canonical objects. d, Final stable anchors retain high shift and dependency ranks across HCC38 and HCC1143, with numeric quantile means annotated per bar and the PFDN5 column highlighted; no covariate-level marking is drawn at this structural layer. e, Per-anchor covariate Total Variation Distance (TVD) matrix across five covariate axes (barcode gem group, UMI-threshold bin, UMI-quantile bin, detected-genes quantile bin, total-signal quantile bin) and the two HCC contexts, for the four final stable anchors. f, Compact anchor claim matrix (conclusion). PFDN5 is the only primary-but-qualified anchor and is covariate-clean; PMF1, PRPF6, and ZNF131 are retained as supporting only because panel e shows they are covariate-exposed despite being structurally stable.



	backbone	shift-excess	separation
shared mean baseline	0.81	0.33	0.35
GEARS formal	0.66	0.33	0.43
Geneformer	0.53	0.75	0.40
scGPT	0.45	0.33	0.29
LM + Geneformer	0.63	0.17	0.33
LM low-rank	0.54	0.58	0.29
LM + scGPT	0.47	0.17	0.26
null	0.50	0.50	0.00

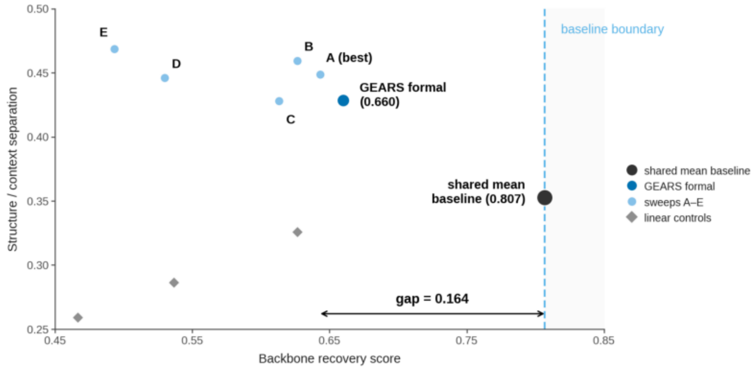
Fig. 3 | Framework element 3: Architecture-aware model adjudication reveals an asymmetric recovery pattern

a, Three adjudication metrics separate recovery modes across entrants. b, Baseline leads backbone recovery, whereas GEARS leads context separation. c, Entrants occupy an asymmetric recovery space. Backbone recovery versus structure-versus-context separation is plotted for the shared-mean baseline, the formal GEARS recipe, foundation-model entrants, and the linear controls.

a **Pre-specified finite-budget GEARS candidates**

Candidate	Epoch	LR	WD	Role
GEARS formal	30	1×10^{-3}	1×10^{-6}	reference recipe
A (best)	30	2×10^{-3}	1×10^{-6}	best sweep candidate
B	20	1×10^{-3}	1×10^{-6}	sweep candidate
C	30	1×10^{-3}	1×10^{-5}	sweep candidate
D	30	5×10^{-4}	1×10^{-6}	sweep candidate
E	40	1×10^{-3}	1×10^{-6}	sweep candidate

b **Prespecified rebuttal candidates do not close the backbone gap**



c **No tested rebuttal candidate closes the backbone gap to the shared-mean baseline**

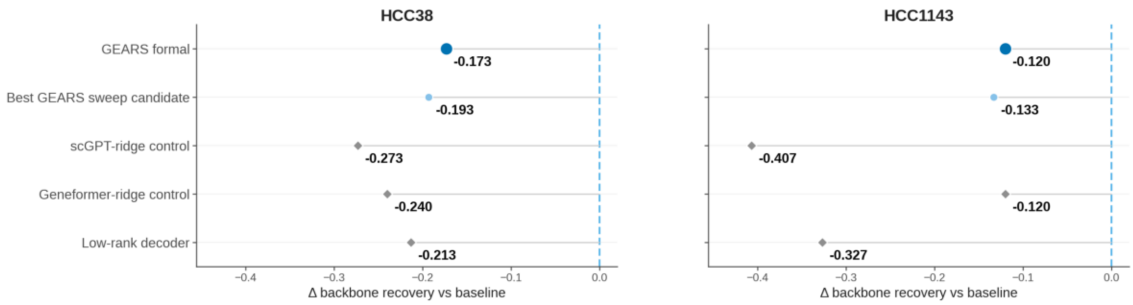


Fig. 4 | Framework element 4: Finite-budget rebuttal tests with a pre-specified stop rule do not close the backbone recovery gap

a, Pre-specified finite-budget GEARS candidates. Six pre-specified recipes were evaluated under an unchanged truth object and scoring system: the formal GEARS recipe and five one-parameter local variants varying epochs, learning rate, or weight decay. b, Pre-specified rebuttal recovery space. The shared-mean baseline, GEARS formal, the five GEARS sweep candidates, and three embedding-based linear controls are placed in the same backbone-recovery versus structure/context-separation space. Diamond markers denote the three linear controls: scGPT-ridge, Geneformer-ridge, and the low-rank decoder. c, Per-context residual backbone gap. GEARS formal, the best GEARS sweep candidate, and the three linear controls are plotted as backbone-recovery differences relative to the shared-mean baseline in HCC38 and HCC1143.

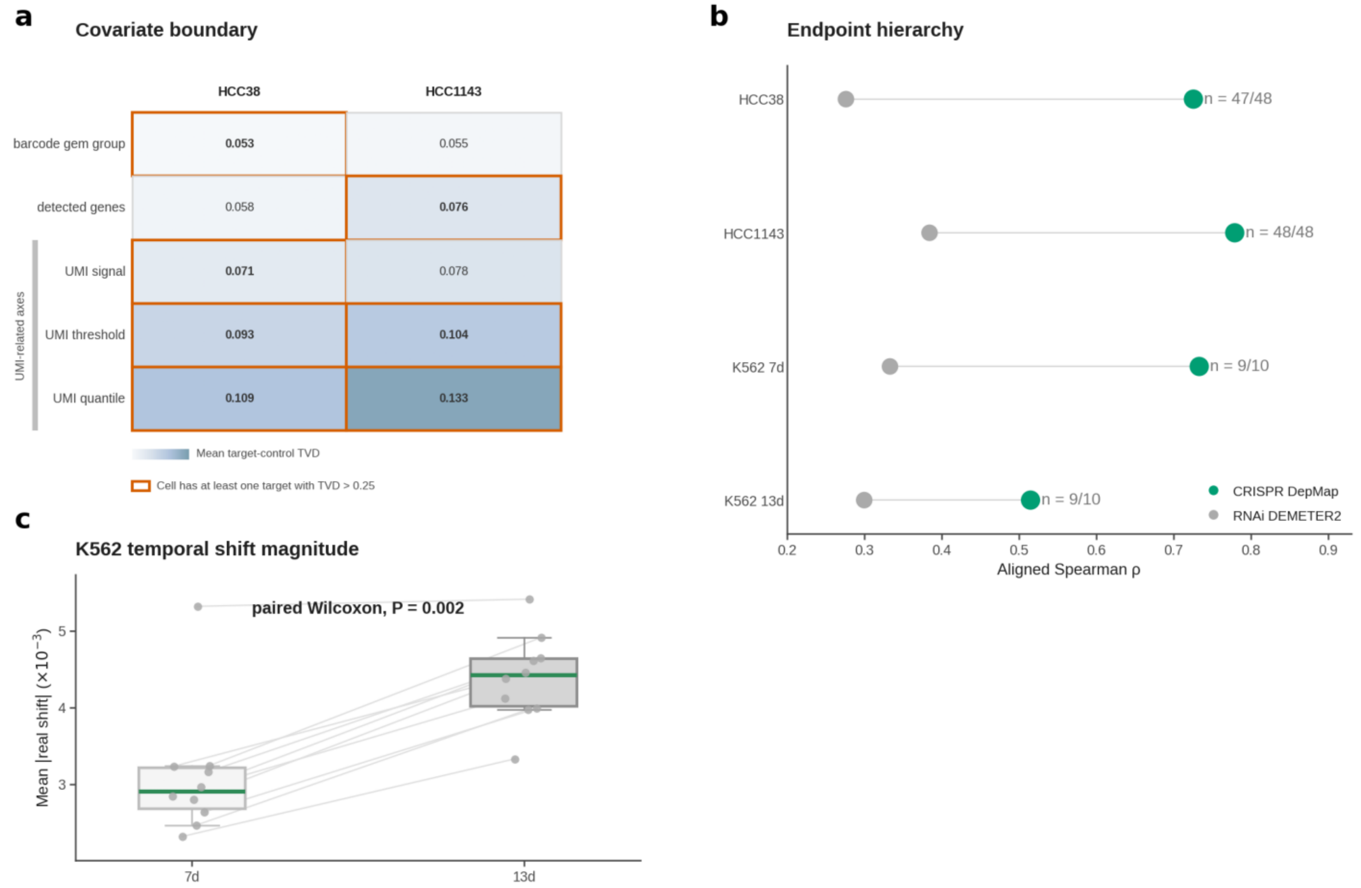
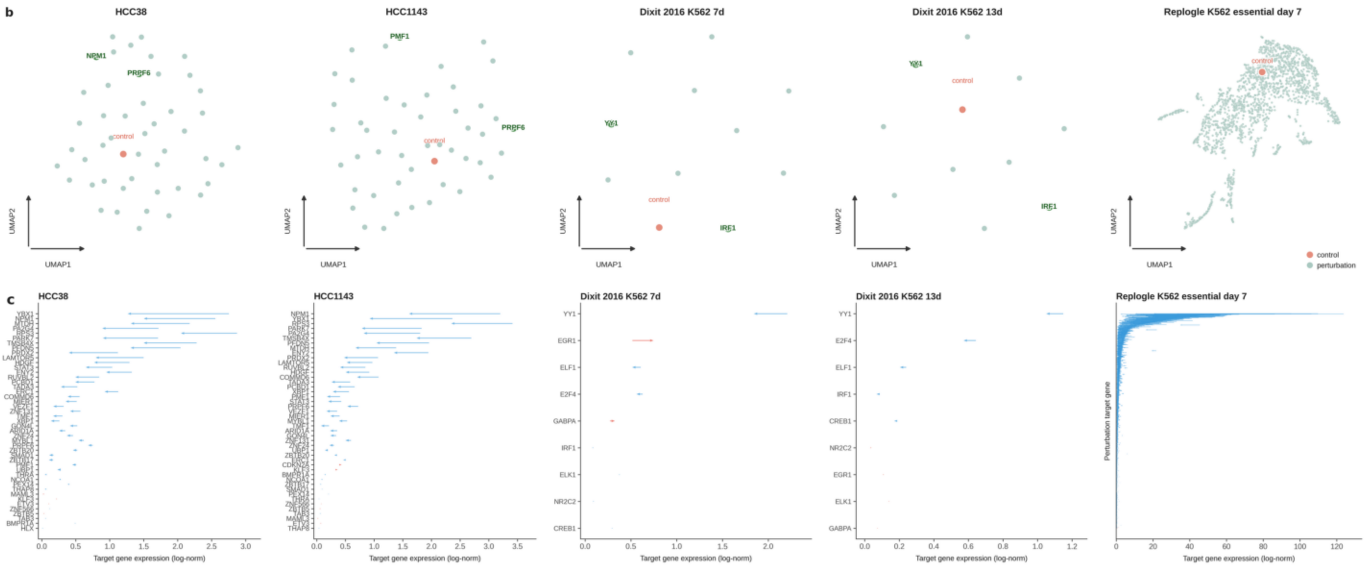


Fig. 5 | Framework elements 6 and 7: Temporal architecture, endpoint hierarchy, and covariate governance define the benchmark claim space

a, Covariate boundary in HCC38 and HCC1143 (Framework element 7). Mean target-control Total Variation Distance (TVD) is shown on a shared color scale for five covariate stratifications (barcode gem group, detected-gene bin, UMI signal, UMI threshold, UMI quantile). b, Endpoint hierarchy across all four bridge contexts (Framework element 6). c, K562 temporal stratification: stronger rank alignment at 7d despite smaller perturbation magnitude (Framework element 6). Paired boxplots summarize per-target mean absolute real shift at 7d and 13d; thin lines link paired targets; jittered points show the per-target values. d, A0/A1/B evidence-tier summary for K562 temporal evidence (Framework element 6).

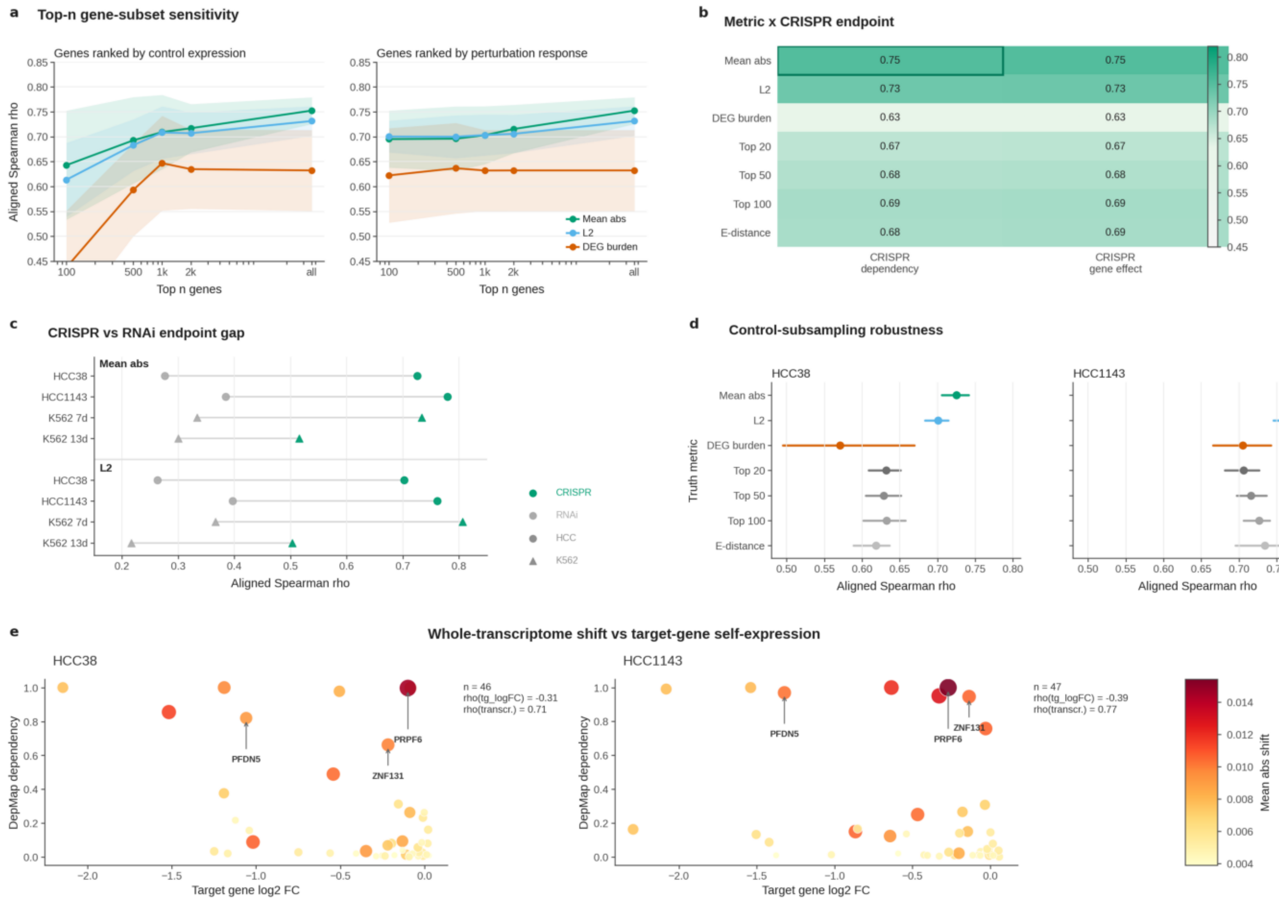
a

Dataset	Size	Benchmark use
Perturbation datasets		
HCC38	36,601 genes x 14,175 cells	47 targets | 1,666 controls
HCC1143	36,601 genes x 11,405 cells	47 targets | 1,325 controls
Dixit 2016 K562 7d	20,824 genes x 28,034 cells	10 targets | 5,381 controls
Dixit 2016 K562 13d	20,824 genes x 15,849 cells	10 targets | 3,491 controls
Replogle K562 essential day 7	8,563 genes x 310,385 cells	2057 targets | 10,691 controls
Endpoint datasets		
DepMap CRISPR dependency (primary endpoint)	1,186 cell lines x 18,435 genes	HCC38 47 / HCC1143 48 / K562 7d 10 / K562 13d 10 / Replogle K562 1,882
DEMETER2 RNAi (sensitivity endpoint)	701 mapped cell lines x 17,309 genes	HCC38 48 / HCC1143 48 / K562 7d 9 / K562 13d 9 / Replogle K562 1,882



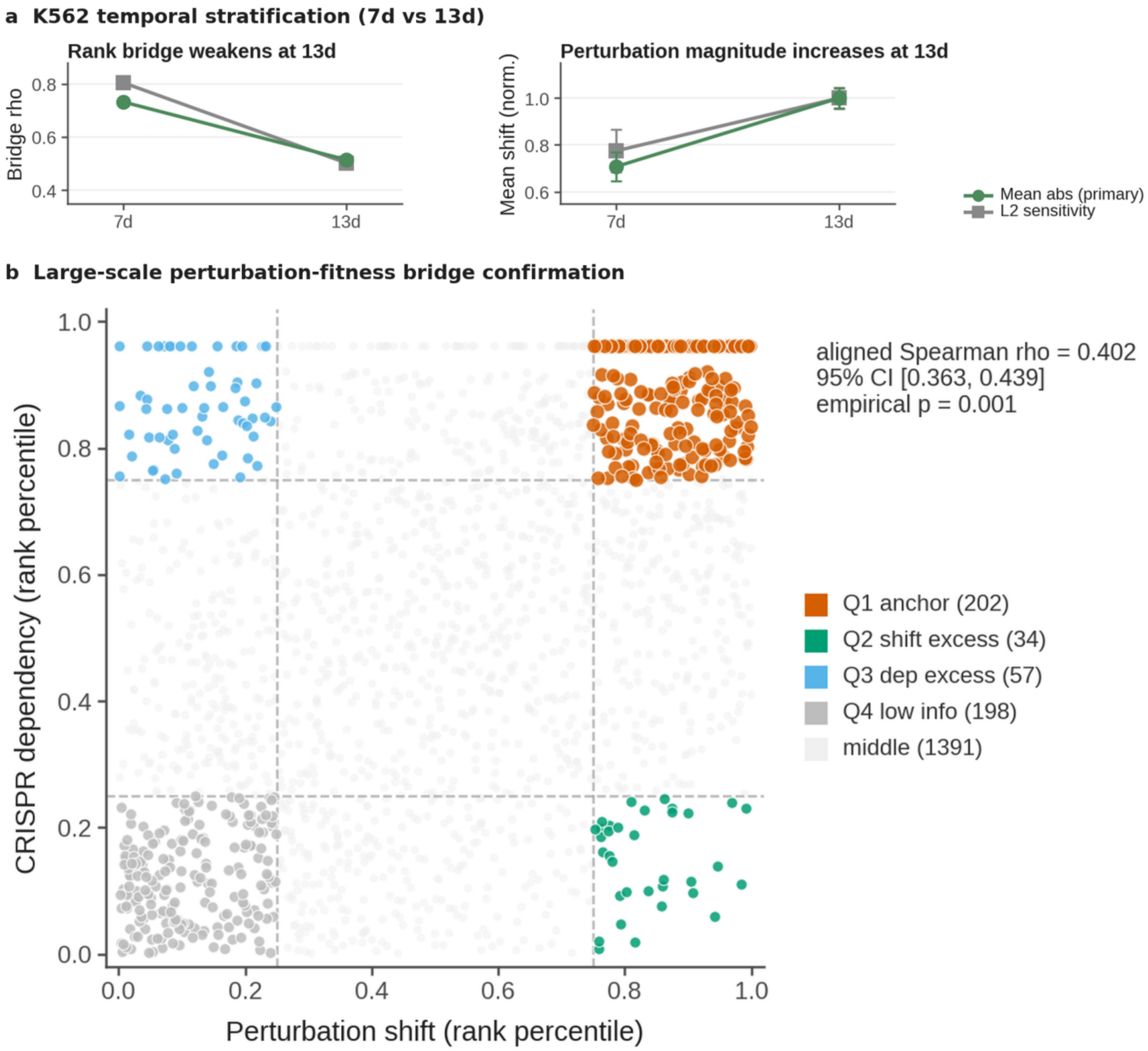
Extended Data Fig. 1 | Dataset familiarization and endpoint inputs

a, Dataset overview table. Five perturbation-expression contexts (HCC38, HCC1143, Dixit 2016 K562 7d, Dixit 2016 K562 13d, and Replogle K562 essential day 7 CRISPRi) and two endpoint datasets (DepMap CRISPR dependency and DEMETER2 RNAi) are summarized with cell-line identity, size, and benchmark use. b, UMAP of perturbation-level mean profiles for HCC38, HCC1143, Dixit 2016 K562 7d, Dixit 2016 K562 13d, and Replogle K562 essential day 7 (five panels, left to right). c, Target-gene expression change for the same five contexts (left to right). Each arrow represents one perturbation target: the base of the arrow indicates the target gene's mean expression in control cells, and the tip indicates its mean expression after perturbation (log-normalized).



Extended Data Fig. 2 | Robustness of perturbation-fitness bridge metric selection

a, Top-n gene-subset sensitivity. b, Metric and CRISPR endpoint robustness. The heat map shows mean aligned Spearman correlation across HCC38 and HCC1143 for candidate truth metrics crossed with the two CRISPR DepMap readouts; the boxed cell marks the retained primary metric and endpoint. c, Endpoint sensitivity. d, Control-subsampling robustness. Aligned Spearman correlation with DepMap CRISPR dependency was recomputed across repeated control-cell subsamples for candidate truth metrics in HCC38 and HCC1143; points show the mean and horizontal intervals show the 2.5th-97.5th percentile range across subsamples. e, Whole-transcriptome shift versus target-gene self-expression in driving the fitness bridge. Each point represents one perturbation target in HCC38 (left sub-panel) or HCC1143 (right sub-panel). The x-axis shows the target gene's own log2 fold-change; the y-axis shows DepMap CRISPR dependency.



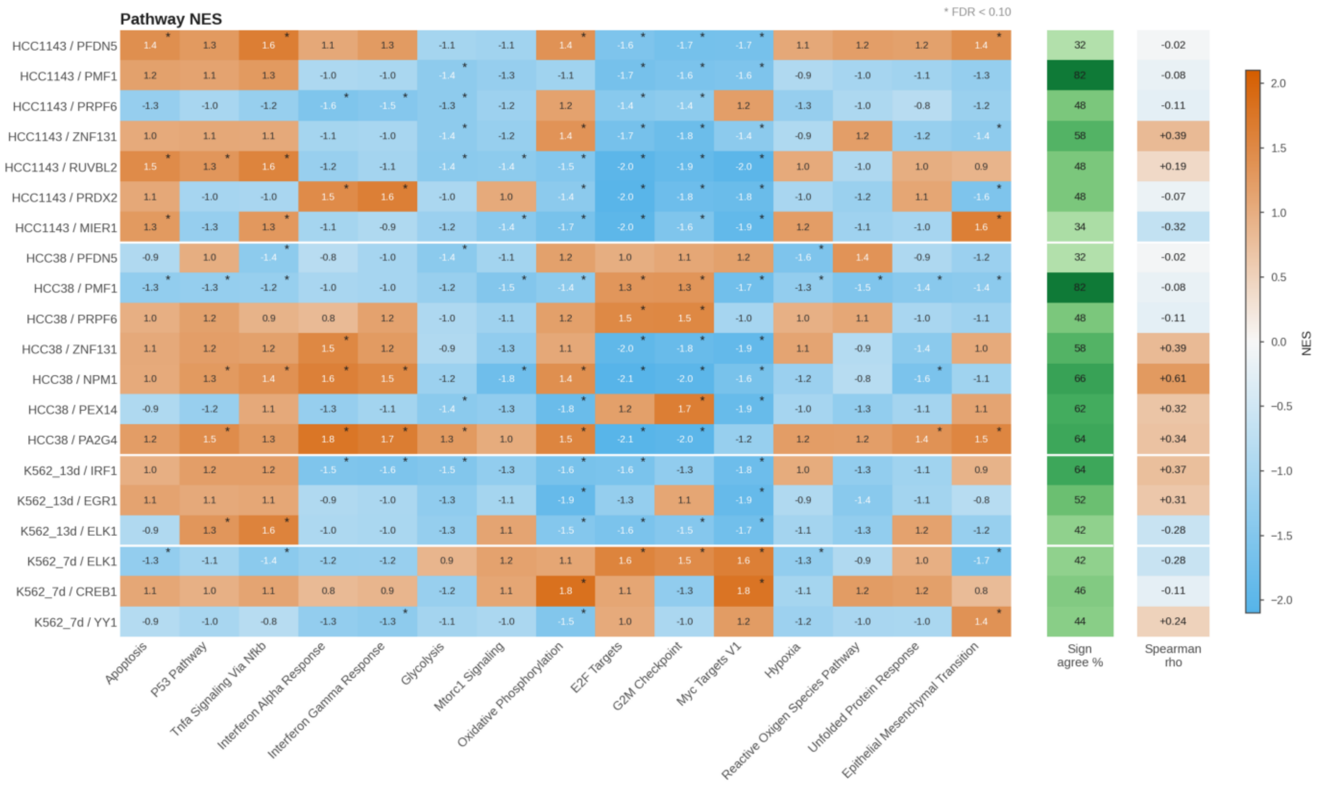
Extended Data Fig. 3 | K562 temporal evidence and large-scale bridge confirmation

a, K562 temporal bridge-magnitude dissociation (GSE90063). Aligned Spearman correlation with CRISPR DepMap dependency and mean perturbation-shift magnitude are shown for the 7d temporal-sensitivity panel and the 13d formal supplementary bridge test. b, Replogle K562 essential day 7 large-scale perturbation-fitness bridge. Scatter of real_shift_mean_abs versus depmap_gene_dependency for 1,882 matched CRISPRi targets, displayed as within-context rank percentiles (0–1 axes, matching the convention of Fig. 1c,d). Dashed lines mark the 25th and 75th percentiles.

Extended Data Fig. 4 | Descriptive axis-level signal space

a, Axis-level signal space comparing dependency signal with transcriptomic shift signal across annotated axes. b, Paired axis R-squared ranking. Axes with the largest value in either signal dimension are displayed with matched shift and dependency R-squared values on the same scale; labels report compact target breadth.

a **Exploratory pathway-level summaries of target-control perturbation responses**



Extended Data Fig. 5 | Exploratory pathway-response polarity heatmap

a, Pathway normalized enrichment score (NES) heat map across the pre-specified display targets in HCC38, HCC1143, K562 7d, and K562 13d. Rows show selected benchmark anchors and high-variance response examples; columns show the pre-specified Hallmark response panel. Asterisks indicate pathway enrichments passing FDR < 0.10. Right-hand summaries report same-target partner-context pathway Spearman correlation and sign-agreement fraction across all Hallmark pathways.